

IRIS UWIZEYIMANA

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EDUCATION

Doctor of Philosophy <i>Computer Engineering</i>	Sep 2022 – Oct 2026 (Expected)
University of Toronto	Toronto, ON, Canada
Thesis: Sustainable Machine Learning Systems: Modeling, Analysis, and Optimizations	
Master of Applied Science <i>Computer Engineering</i>	Sep 2020 – May 2022
University of Toronto	Toronto, ON, Canada
Thesis: Compiler Optimization for Deep Neural Networks.	
Bachelor of Science <i>Computer Engineering (Summa Cum Laude)</i>	Aug 2016 – May 2020
Illinois Institute of Technology	Chicago, IL, USA

RESEARCH EXPERIENCE

PhD Research	Sep 2022 – Present
University of Toronto	Toronto, ON, Canada
– <u>A Heterogenous Mapping of A Speech Neuroprosthesis Application (ISPASS 2026)</u> * Designed a pipelined CPU/GPU/NPU mapping for a speech neuroprosthesis application on a AMD Ryzen heterogeneous SoC. * Profiled the full application to identify performance bottlenecks and built an early roofline model to estimate potential speedups. * Implemented a GPU mapping of the Viterbi graph traversal stage of the application using AMD HIP. * Mapped the RNN stage to an AMD Strix Point NPU using MLIR-IRON and AMD AIE libraries. – <u>A Language Model Latency and Energy Predictor (IISWC 2026 – Submitted)</u> * Built a database of latency and power profiles from commonly used PyTorch and Hugging Face operators. * Designed a framework that uses a dataflow-graph abstraction and the per operator profiles to estimate the end-to-end latency and energy of transformer-based models. * Studied the impact of early exit on model performance under different exit distributions. – <u>A Carbon-Aware Server Replacement Policy Maker (ISPASS 2025 Extended Abstract)</u> * Developed an analytical model for optimizing server replacement decisions under embodied and operational carbon constraints. * Quantified the tradeoff between manufacturing emissions from new servers and operational emissions from aging hardware. * Evaluated when extending server lifetime can reduce total data center carbon emissions.	
Master's Research	Sep 2020 – May 2022
University of Toronto	Toronto, ON, Canada
– <u>Tackling Resource Utilization in Deep Neural Network Accelerators</u> * Automated DNN accelerator design-space exploration using the MAESTRO analytical model to optimize tiling, unrolling, and dataflow parameters. * Developed scheduling algorithms for concurrent mapping of CNN layers on a single accelerator * Used a Bayesian Optimization algorithm for the dataflow DSE and a swarm-based heuristic algorithm for resource distribution.	

WORK EXPERIENCE

Software Engineering Intern	May 2022 – Aug 2022
Cerebras Systems	
– Worked on compiler optimizations of CNN workloads targeting the Cerebras Wafer-Scale Engine, helping in the developments of a color-based routing algorithm for inter-core data transfers.	

SKILLS

- **Languages:** English, French, Kinyarwanda.
- **Programming Languages:** Python, C++, Verilog, MATLAB, (RISC-V, ARM and MIPS) assembly language.
- **Frameworks:** Pytorch, CUDA, HIP, MLIR-IRON